Solar ESS Co. Ltd. · บริษัท โซลาร์ เอสส์ จำกัด
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PROPOSAL · ON-GRID SOLAR ROOFTOP

2 × 15 kW Parallel On-Grid Solar Rooftop System

Prepared for **a subdistrict municipality** , office building · 12 October 2026

TURNKEY PRICE

— THB before VAT

— THB including VAT · second inverter supplied free of charge

— THB per watt
31,200 W installed

31.2 kWp
48 × LONGi Hi-MO X10 650 W

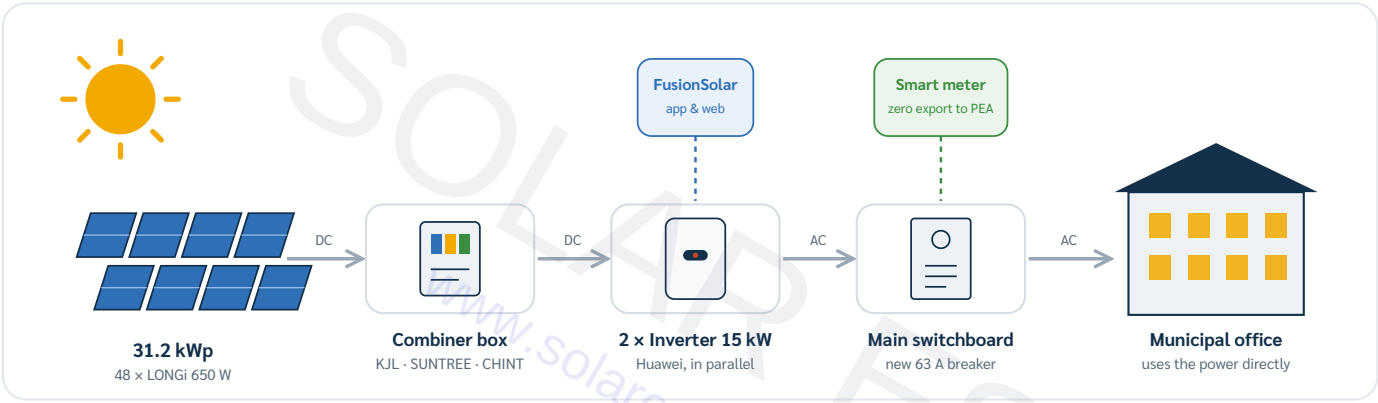
2 × 15 kW
Huawei SUN2000-15KTL-M5 in parallel

41,700 kWh
production in year 1

4.2 years
payback on — THB

2.62 M THB
net saving over 25 years

≈ 17 t CO_2
avoided every year



Solar ESS will survey, design, supply, install and connect two 15 kW Huawei inverters in parallel, handle the PEA application, and maintain the system for two years. The second inverter is supplied free of charge. Both inverters feed the office directly, and one smart meter keeps export to the PEA grid at zero.

1 About Solar ESS

Solar ESS Co. Ltd. (บริษัท โซลาร์ เอสส์ จำกัด) · "Solar Energy for Sustainable Solution" · Tax ID 0575566002051

We design and install turnkey solar systems from our head office in Chiang Rai, with branches serving the south of Thailand.

Offices
Head office Chiang Rai · branches in Krabi, Koh Samui (Surat Thani) and Khanom (Nakhon Si Thammarat)

Services
Survey, design, supply, installation, PEA application with a signed SLD, cleaning and maintenance

Systems
On-grid, hybrid and off-grid, from homes to commercial and institutional buildings

Equipment
Huawei inverters on most of our sites; LONGi panels; SUNTREE, CHINT and ABB protection; KJL enclosures; Yazaki and LINK cable

Solar ESS Working Group	Watcharote Iamklam	Anucha Intongead	Artitaya Khuanphet
Chief Executive Officer	Chief Technical Officer · licensed electrical engineer (ภฟท. 44981)	Chief Operation Officer	Chief Legal Officer

Reference sites can be viewed on our website, www.solaress.org.

2 Proposed system

A 31.2 kWp array on two Huawei 15 kW inverters in parallel, joined in one combiner box and connected to the existing switchboard with zero export.

PV array
48 × LONGi Hi-MO X10 650 W (LR7-72HVH-650M), 4 strings of 12 = 31.2 kWp

Inverter
2 × Huawei SUN2000-15KTL-M5, 15 kW each, 3-phase 400 V, 2 MPPT each; INV-02 supplied free of charge

Grid connection
Each inverter on its own 32 A CHINT RCBO, joined on a 63 A busbar and ABB main breaker, then a new 63 A CHINT breaker in the MDB

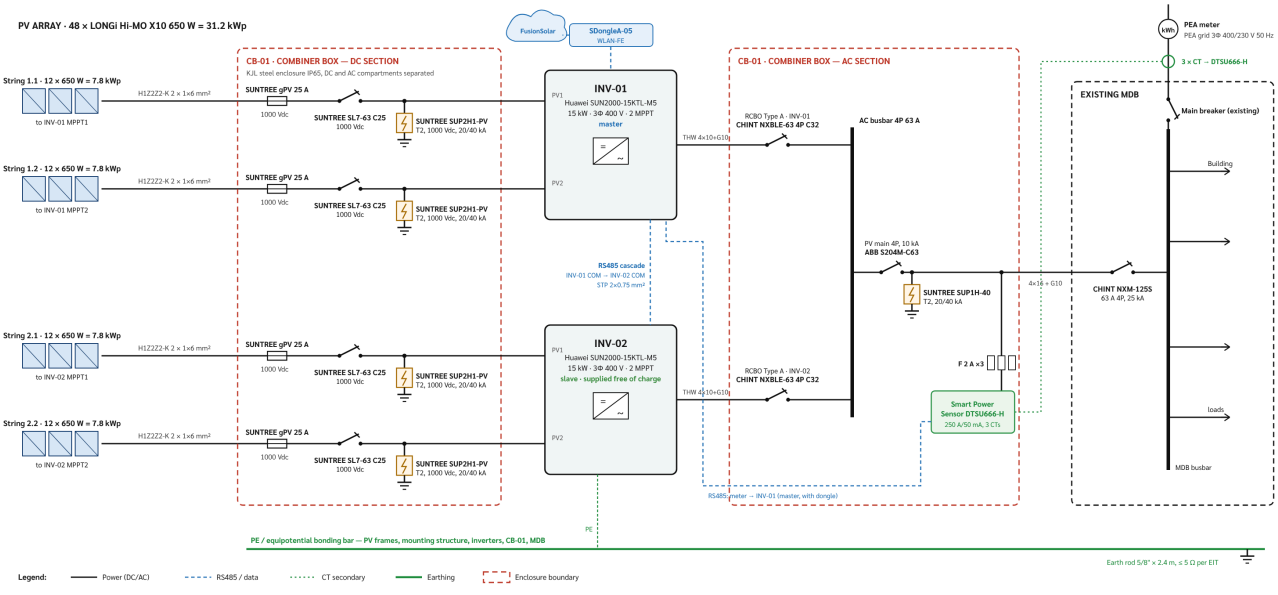
Monitoring
Huawei SDongleA-05 on INV-01 (master); INV-02 cascaded on RS485; one FusionSolar plant

Combiner box
KJL steel IP65 with SUNTREE DC and CHINT AC protection

Mounting and earthing
Aluminium rails, stainless fittings, frames bonded to an earth rod

Single line diagram

Each of the 4 strings runs through its own SUNTREE fuse, DC breaker and surge protector to one MPPT. Each inverter has its own CHINT RCBO; the two join on a 63 A busbar and pass the ABB main breaker and AC surge protector to a 63 A breaker in the MDB. The DTSU666-H meter on INV-01 holds export at zero for both inverters.

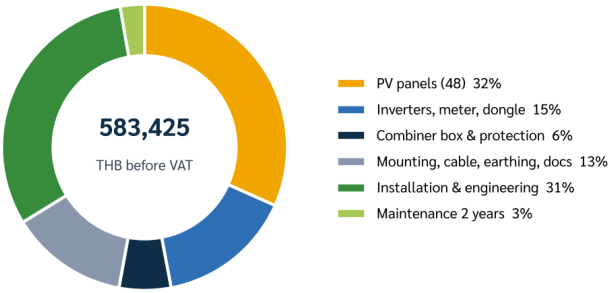


3 Bill of materials and price

– THB before VAT, – THB per watt: – THB materials and – THB labour. The second inverter is supplied free of charge. VAT 7% applies to materials only.

#	Item	Brand / model	Qty	Unit (THB)	Amount (THB)
PV array					
1	PV module 650 W, HPBC 2.0, delivered to site	LONGi Hi-MO X10 LR7-72HVH-650M	48	–	–
Inverters and monitoring					
2	On-grid inverter 15 kW, 3-phase, 2 MPPT (INV-01, master)	Huawei SUN2000-15KTL-M5	1	–	–
3	On-grid inverter 15 kW, 3-phase, 2 MPPT (INV-02, slave), supplied free of charge	Huawei SUN2000-15KTL-M5	1	–	–
4	Monitoring dongle (WLAN-FE), on INV-01	Huawei SDongleA-05	1	–	–
5	Smart meter 3-phase with 3 CTs, zero export for both inverters	Huawei DTSU666-H	1	–	–
Combiner box and protection					
6	Combiner box, steel IP65, lockable, DC and AC compartments	KJL	1	–	–
7	DC fuse holder 1100 Vdc + gPV fuse 25 A	SUNTREE SRD-30 + SRF-30	8	–	–
8	DC MCB C25, 1000 Vdc, 1 per MPPT	SUNTREE SL7-63 4P	4	–	–
9	DC SPD Type II, 1000 Vdc, 20/40 kA, 1 per MPPT	SUNTREE SUP2H1-PV 3P	4	–	–
10	RCBO 4P C32, Type A, 100 mA, 1 per inverter	CHINT NXBLE-63	2	–	–
11	Copper busbar 4P 100 A with PE bar	KJL	1	–	–
12	MCB 4P 63 A, 10 kA, PV main	ABB S204M-C63	1	–	–
13	AC SPD Type II 4P, 385 Vac + backup MCB	SUNTREE SUP1H-40 + SCB8-63	1	–	–
14	Meter fuse holder 1P + 2 A fuse	SUNTREE SRD-30	3	–	–
15	Terminals, wiring duct, DIN rail, labels	KJL	1 lot	–	–
16	MCCB 4P 63 A, 25 kA, PV feeder in the MDB	CHINT NXM-125S/4P	1	–	–
17	RS485 cable STP 2×0.75 mm², INV-01 → INV-02 and meter → INV-01	–	1 lot	–	–
Mounting, cable, earthing and documents					
18	Aluminium rails, L-feet, clamps, stainless bolts	Aluminium + SUS304	48 panels	–	–
19	PV cable H1Z2Z2-K 1×6 mm², 1.5 kV DC	LINK	400 m	–	–
20	MC4 connector pair, 1500 Vdc	SUNTREE	12	–	–
21	AC cable THW 1×10 mm², inverters → combiner box, and earth	Thai Yazaki	130 m	–	–
22	AC cable THW 1×16 mm², combiner box → MDB	Thai Yazaki	120 m	–	–
23	Earth rods 5/8" × 2.4 m, earth cable, clamps and bonding	KUMWELL + Thai Yazaki	1 lot	–	–
24	Ground test box	KUMWELL GYPTB	1	–	–
25	Metal trunking with cover, conduit, fittings	KJL	1 lot	–	–
26	As-built drawings (A3 × 2, DWG + PDF on 3 flash drives), Thai manual × 2	Solar ESS	1 lot	–	–
Labour (not subject to VAT)					
27	Installation, testing and commissioning, 31,200 W at – THB/W	Solar ESS	31,200 W	–	–
28	Engineering, shop drawings, PEA application and SLD signed by a licensed engineer, parallel and RS485 set-up	Solar ESS	1 lot	–	–
29	Maintenance for 2 years, panel cleaning once a year	Solar ESS	2 years	–	–

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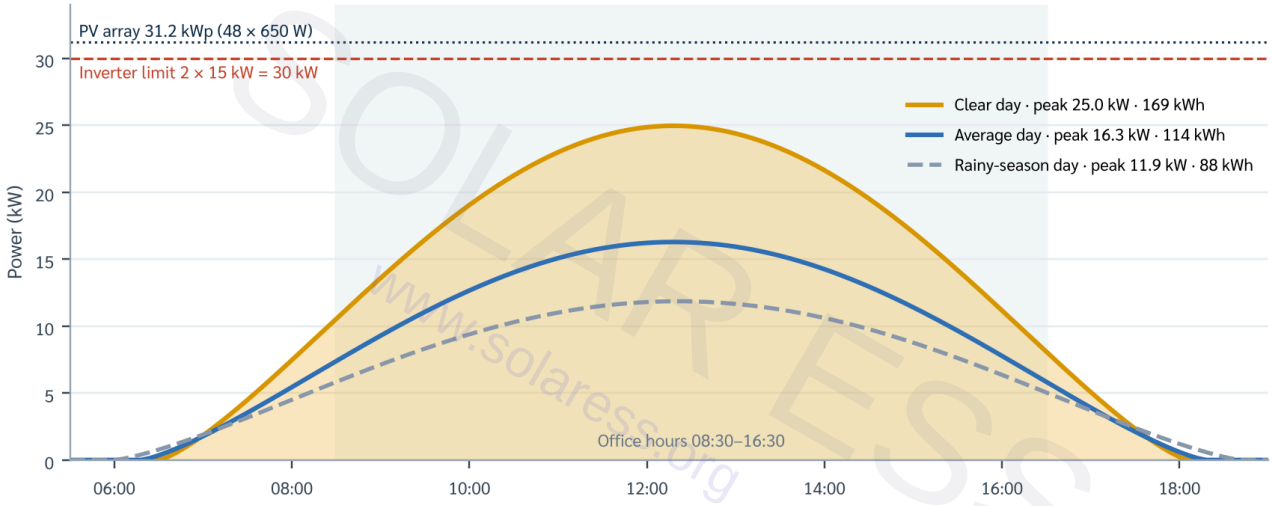


Materials	–
Labour (no VAT)	–
Total before VAT · – THB/W	–
VAT 7% on materials	–
Grand total including VAT	–

4 Energy production

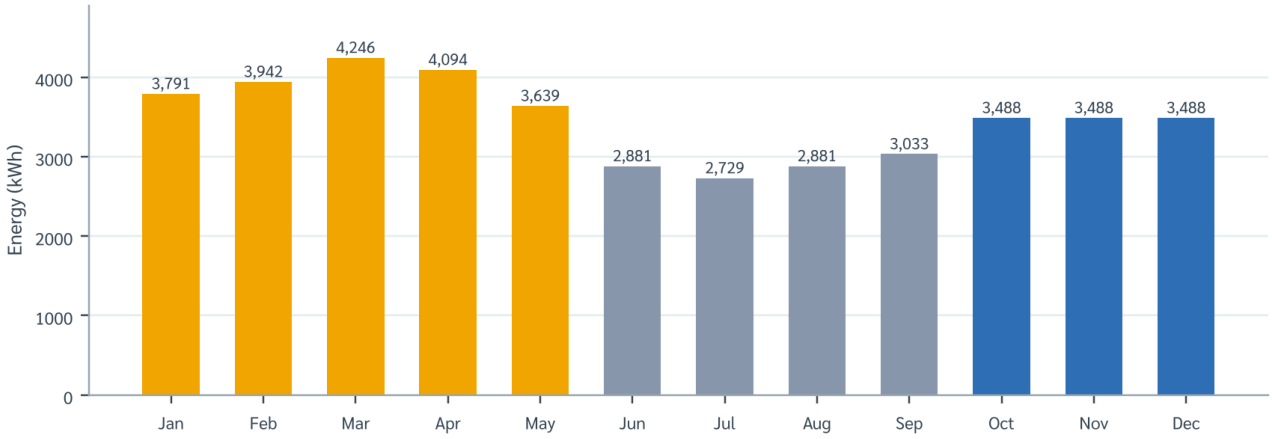
About 41,700 kWh in year 1. On a clear day the two inverters together peak at about 25 kW, below their combined 30 kW limit, so nothing is clipped.

Through the day, against panel and inverter capacity



Most output falls inside office hours, when the building uses it directly. DC-to-AC ratio 31.2 kWp : 30 kW = 1.04, 15.6 kWp on each inverter.

By month, year 1



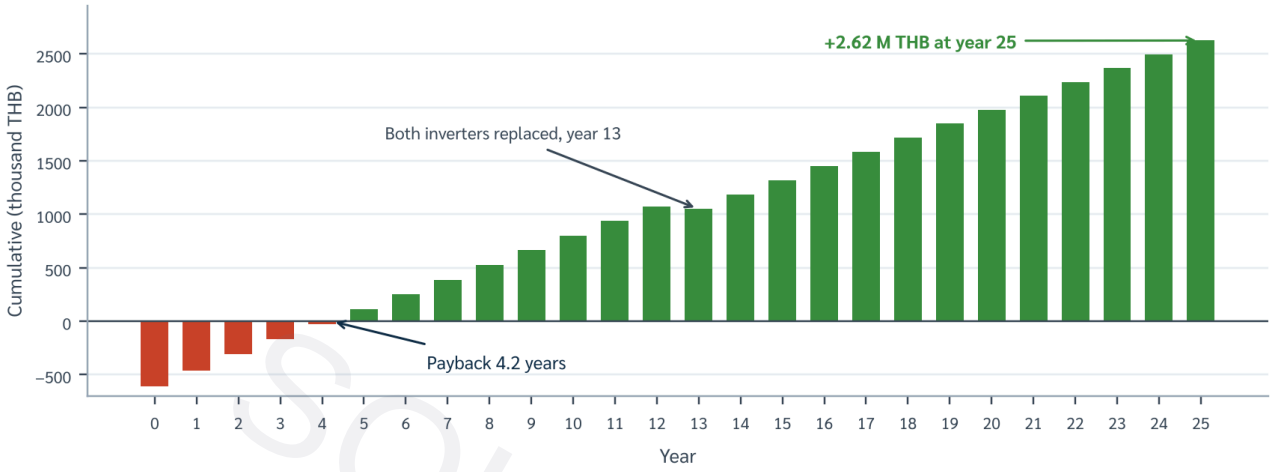
March and April produce most (about 4,100–4,250 kWh each); July produces least (about 2,730 kWh) because of monsoon cloud. Typical Chiang Rai profile scaled to 1,350 kWh per kWp, less 1% first-year degradation.

5 Return on investment

The – THB investment pays back in about 4.2 years and returns 2.62 million THB over 25 years.

4.2 yrs	22.8%	150,116 THB	≈ 17 t
payback	IRR over 25 years	saving in year 1	CO_2 avoided a year

Cumulative cash flow, base case



Assumptions

Investment	– THB incl. VAT (not reclaimable)
Production	1,350 kWh/kWp; –1% year 1, then –0.4%/yr
Used in office	80% (rest held back by zero export)
Electricity price	4.50 THB/kWh incl. Ft and VAT, flat
Running costs	8,000 THB/yr from year 3; both inverters replaced (159,000 THB) in year 13

Sensitivity

Case	Payback	25-yr saving	IRR
60% used	5.7 yrs	1,730,151	16.1%
80% used (base)	**4.2 yrs**	**2,624,690**	**22.8%**
100% used	3.3 yrs	3,519,230	29.2%
80%, tariff +2%/yr	4.1 yrs	3,612,086	24.9%

The share of output the office uses is the figure to confirm: with 31.2 kWp, midday output may exceed the office load, and the office closes at weekends, so a 15-minute load profile from the MDB would firm up the base case. CO_2 at the Thai grid factor of about 0.5 kg/kWh.

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6 Scope of work and schedule

Delivered turnkey within 90 days of contract signing, as the TOR requires; site work takes about 30 days.

1. **Survey and design**
   Roof, structure load and MDB survey; calculations and shop drawings signed by a licensed engineer within 7 working days
2. **Supply**
   All equipment new and unused, with manufacturer certificates
3. **Installation**
   Rails, panels, combiner box, inverter, trunking and earthing; roof fixings sealed against leaks
4. **Connection**
   New 63 A breaker in the existing MDB; any shutdown agreed 7 working days ahead, 09:00–16:00
5. **Testing**
   Commissioning to IEC 62446-1, parallel and RS485 set-up, zero-export setting and FusionSolar monitoring
6. **PEA approval**
   Grid-connection application handled by Solar ESS until approved for use
7. **Handover**
   As-built drawings, Thai O&M manual and on-site training for municipal staff

7 Warranty and maintenance

All warranties meet or exceed clause 11 of the TOR.

PV modules, product
At least 10 years, per the LONGi warranty certificate

PV modules, power output
At least 25 years linear, not less than 80% output

Inverters
10 years each, claims handled directly by Huawei Thailand, including the free INV-02

Workmanship and materials
2 years from handover

Repair response
On site within 7 days of notice; repaired part re-warranted for 2 years

Maintenance
Cleaning and electrical safety check once a year for 2 years, included

8 Terms

- **Payment:** one payment after completion, PEA approval for use and acceptance by the inspection committee (TOR clause 9).
- **Price:** Thai baht; VAT 7% on materials, labour not subject to VAT; valid until 11 November 2026.
- **Included:** PEA connection application, signed SLD, shop drawings, as-built drawings, manual and training.
- **Not included:** roof repairs beyond the panel fixings, MDB replacement, and an application to sell power to PEA (2,140 THB if requested).
- **Delivery:** within 90 days of contract signing.

Proposed by
Solar ESS Working Group
Chief Executive Officer, Solar ESS Co. Ltd.
Date _____

Accepted by
a subdistrict municipality

Date _____